A STORY SERIES FOR YOUNG READERS

TALES UNDER THE MOON

✦ ✦ ✦

VOLUME 1

Chú Cuội, Chị Hằng and the Forgotten Friends

For readers aged 5–14 and the grown-ups who read along

LIGHT OF MEMORIES

LIGHT OF
MEMORIES

TALES UNDER THE MOON

Volume 1

Chú Cuội, Chị Hằng

and the Forgotten Friends

Authors

LIGHT OF MEMORIES

Lê Phạm Thế Vũ

Nguyễn Đức Mạnh

Nguyễn Bảo An

Nguyễn Thị Thanh An

Cao Quốc Khánh

Vũ Lê Đức Minh

About this book

This book was developed with the support of Claude AI (Anthropic), SunoAI and OpenAI Codex. The authors are responsible for the final content.

It was written for the NASA International Space Apps Challenge 2026 and is not sponsored or endorsed by NASA. The spacecraft and events mentioned are real; the characters and conversations are invented. The “Did you know?” section at the end helps you tell real from story.

Contents

Introduction **3**

Part 1: Awakening **4**

Part 2: The Forgotten Friends **6**

Chapter 1: Round Eye 7

Chapter 2: Twinkle 12

Chapter 3: Sharp Ear 18

Chapter 4: Mesh Wheels 24

Chapter 5: Sky Watcher 29

Part 3: The Light Returns **34**

Afterword **36**

Reader's Notebook **37**

Did you know? Round Eye 37

Did you know? Twinkle 39

Did you know? Sharp Ear 41

Did you know? Mesh Wheels 43

Did you know? Sky Watcher 46

Little Dictionary 48

Questions to Talk About 50

Sources and AI Used **52**

Acknowledgements **54**

Introduction

Dear young readers,

Have you ever looked up at the Moon on a full-moon night and wondered, “Is anyone up there?”

In Vietnamese folklore, a woodcutter named Cuội sits beneath a banyan tree on the Moon, and a gentle lady named Hằng lives in the Moon Palace. But if you had a very big telescope, you would find that the Moon has other residents too: small machines that people sent there long ago. They worked hard. They took pictures, dug in the dust, listened and measured. Then one day they quietly stayed behind, and nobody called their names any more.

This book tells how Cuội set out to find Hằng, and about the forgotten friends he meets along the way. Each friend has a story. Read slowly, and try to remember every name.

What we want to share is simple: science is lovable, and everything people make, whether a small machine or a plastic bag, deserves to be remembered and taken responsibility for.

At the end of the book you will find a section called **“Did you know?”**. There you can learn which friends are real, what they did, and which parts are only in the story.

The Light of memories team

Part 1: Awakening

A Full-Moon Night

On the night of the Mid-Autumn full moon, the Moon was round and bright as a silver plate. Little An sat on the porch, holding a star lantern, gazing up at the sky.

> ***[Illustration suggestion]*** *A porch on Mid-Autumn night: little An sits beside Grandma, a five-pointed star lantern in her hands; a huge round moon rises behind the roof.*

- Grandma, Cuội has been sitting under that banyan tree for such a long time. Isn't he sad?
- Long ago he wasn't sad, dear, because Hằng was beside him. But I've heard that one night Cuội woke up… and Hằng was no longer there.

The lantern in An's hands flickered softly. A light breeze crossed the porch. And that is where the story begins.

· · ·

The Banyan Tree

Long, long ago there lived a woodcutter named Cuội. He lived beside a great banyan tree and cherished it like an old friend. But people say that one day someone carelessly poured a bucket of dirty water at the foot of the tree. The banyan shuddered, pulled up its roots and slowly rose into the sky, carrying Cuội, who was hugging its trunk. They flew and flew until they landed on the Moon.

There Cuội met Hằng. She was gentle as moonlight; wherever she walked, the air smelled of cinnamon blossoms and soft music drifted by. From then on Cuội sat beneath the banyan and told her stories. Hằng smiled, and the moonlight grew a little brighter.

One day Cuội fell asleep. He slept a very long sleep, so long that the banyan shed its leaves and grew them back again many, many times.

When Cuội woke, the Moon was strangely quiet. The stone seat beside him was still there, but Hằng was not. There was no scent of cinnamon, and no music.

- Hằng! — Cuội called.

Only an echo answered.

> ***[Illustration suggestion]*** *Cuội sits up beneath the banyan and looks at the empty stone seat beside him. Pale moonlight, long shadows across the grey dust.*

Cuội got up and walked around the tree. In the grey dust he saw things that had never been there before: **footprints**, with every ridge of the boot soles pressed clear; long **wheel tracks** stretched across the plain; and in the distance, some metal shapes lay still, as if asleep. They were neither rocks nor trees.

- Who came here while I was sleeping? — Cuội whispered.

The banyan rustled its leaves. Its voice was low and slow, like the night wind:

- She has not gone far, child. But the Moon has changed. While you slept, new friends arrived. They have seen more than you imagine. Go and find them.

From a low branch the banyan lowered a **five-pointed star lantern**. It was dark, but when Cuội took it, it felt as warm as a glowing ember.

- Each friend you find will light one point of the star. When the star shines fully, the moonlight will show you where she is.
- But how will I know who is a friend? — Cuội asked.
- Whoever is waiting for you to remember their name, that one is a friend.

> ***[Illustration suggestion]*** *The ancient banyan with its great roots; the dark five-pointed star lantern is placed in Cuội's hands.*

Cuội remembered the bucket of dirty water, and how one careless act had carried the banyan and himself to this place. He told himself: this time, I will do it right.

He looked at the dark lantern, then at the strange footprints in the grey dust.

- All right — he said — let's go.

And Cuội set off, following the first footprints ...

Part 2: The Forgotten Friends

The banyan was right: there were friends waiting for Cuội on the Moon. They had arrived in different years, each with a different job and a different story. They were not alike, but every one of them had once worked hard, and every one of them wished for the same thing: **not to be forgotten**.

Cuội will meet them one by one, in the order they came to the Moon. Let us go with Cuội to meet the friends ...

> ***[Illustration suggestion]*** *Cuội, surrounded by images of the forgotten friends, with circles arranged in the shape of a star lantern and faded pictures.*

Chapter 1: Round Eye

Cuội followed the footprints. The farther he went, the more footprints there were, until they gathered around a wide, shallow crater shaped like an upturned dish. At the bottom sat a three-legged machine, silent, covered in a thin layer of dust.

> ***[Illustration suggestion]*** *Cuội stands on the rim of the crater, looking down at Surveyor 3 sitting quietly on three legs in the grey dust; footprints and wheel tracks lead toward the crater.*

Its body was rough and silvery grey. Here and there were empty places, like missing puzzle pieces. Cuội hesitated, then stepped down, and soft puffs of dust rose around his feet.

- Hello? — Cuội called softly.

A tiny light on the hull blinked. Then a slow, slightly rusty voice, like someone who hasn't spoken for a very long time, said:

- Is… is someone really there? It has been so long since anyone talked to me.
- I'm Cuội. Who are you?
- My name is long and hard to remember: Surveyor 3. But please call me **Round Eye**. I once had a very big eye. It could see very far.

Cuội looked around Round Eye's body. There was an empty place on top, and another empty place on one side, as if someone had taken away the most precious things.

- Where did that eye go, Round Eye?
- That is a long story. You first: where are you going, to have come all the way here?

Cuội held up the dark star lantern and told Round Eye about Hằng, about the banyan, and about the five points of the star.

- That light… — Round Eye whispered. — Once, long ago, I saw a soft glow far away, and then it vanished. I don't know whether it was Hằng. But I would love to help you. Only… I need your help first.
- With what?

- I am about to forget my own story. If you help me remember my three most important pictures, I will light the first point of your star.

· · ·

Three Pictures to Remember

Round Eye said that long ago, from far away on Earth, people had sent him here. He was an **uncrewed machine**, built to find out whether the Moon was solid enough for those who would come after him to stand on.

Round Eye fell silent for a moment. A faint image appeared in the air like an old dream: a three-legged machine, complete, with a big eye on top and a small arm with a scoop on its end.

- This is how I used to be — Round Eye said. — The eye on top is a camera. The arm with the little scoop digs the soil. I used those two things to see and to test the ground.

> ***[Illustration suggestion]*** *Round Eye's memory: Surveyor 3, whole again, with its camera on top and its soil-scooping arm, glowing faintly like a dream.*

Round Eye handed Cuội a **memory shovel**, which could only dig up things that time's dust had buried. Cuội took it and, together with Round Eye, dug down into the dust.

The First Picture: My Foot

The dust parted, and a picture slowly appeared in it: a round footpad of the spacecraft, pressed deep into the ground.

- This is when I landed — Round Eye said. — But I didn't land only once. I touched down, bounced up, touched down again, bounced up again, and only then did I come to rest in this crater.
- Weren't you scared?
- A little. But I thought: however many times I bounce, I still have my work to do.

> ***[Illustration suggestion]*** *A footpad of Surveyor 3 resting on Moon dust, beside the marks left by its bounces.*

The Second Picture: Little Trenches

Cuội dug on. The second picture appeared: small trenches lying side by side in the ground.

- These are the places I dug — Round Eye said. — I used my arm with the scoop to scrape, to dig, to press, to find out whether the soil here was soft or hard. That's how people on Earth learned that Moon dust was firm enough to stand on.
- So you dug the way for those who came after!
- Yes. And there is something odd about this picture: I didn't take it. It was taken in 1969 by visitors who came to see me, but the trenches in it are the ones I dug in 1967.

> ***[Illustration suggestion]*** *Small trenches in the Moon dust, with a caption: "Photographed in 1969, but the trenches were dug in 1967."*

The Third Picture: The Visitors

Cuội dug to the last layer. A picture rose up, warmer than the first two: two people in thick white suits and round helmets standing beside a three-legged machine.

- That was November 1969 — Round Eye said, his voice suddenly brightening. — Two people came to visit me. One was called Pete, the other Alan. They were astronauts.
- Why did they come?
- To meet me. They walked all the way to the crater's edge. They looked at me, touched me, and then gently took off my eye and some of my other parts and carried them back to Earth, so scientists could see how I had endured the Moon. The rest of me stayed here.
- Were you sad?
- At first, yes. But now I think: one part of me went home, and one part stayed to keep the Moon from feeling so empty. And as long as someone remembers me, I am never truly alone.

> ***[Illustration suggestion]*** *Two astronauts in spacesuits stand beside Surveyor 3 at the crater's edge; one reaches out as if touching the machine.*

• • •

Round Eye's Diary

Three pictures lay in Cuội's hands. Round Eye said:

- Now help me put them in the right order, so my story doesn't get mixed up.

Cuội arranged them: first the footpad, because that was when Round Eye arrived. Then the little trenches, because that was when Round Eye worked. Last, the two visitors, because that was when someone came to see him. The three pictures joined into a small diary, open in the grey dust.

Round Eye read each page aloud, slowly and warmly.

- My name is Surveyor 3. I came to the Moon in April 1967. I sent back more than six thousand pictures and dug the first trenches to test the soil. I was remembered.

When Round Eye finished, something wonderful happened. Cuội and Round Eye said together, very softly:

Please don't forget my name.

I remember you now. I will tell your story.

The star lantern trembled in Cuội's hands. A tiny ray of light flew from Round Eye's body into the lantern, and the **first point of the star** lit up, golden as candlelight.

> ***[Illustration suggestion]*** *Cuội holds up the star lantern; its first point glows gold, casting warm light on Surveyor 3.*

- Thank you, Cuội — Round Eye said. — Now let me show you the way.

The little light on Round Eye's hull turned toward a distant direction. There, in the grey night, a small point of light was blinking, bright and dark, bright and dark, as if winking.

- That is a friend of mine — Round Eye said. — She loves to play with light. Go and meet her, Cuội. And tell her my story too.

Cuội nodded. He waved goodbye to Round Eye and walked toward the blinking light, following the footprints, in the very gentle wind of a moonlit night.

Chapter 2: Twinkle

Cuội followed the blinking point of light. He passed through lands so far away that, if you told people about them, they would think it was a fairy tale. Then the grey dust under his feet grew finer and flatter, and Cuội came to a wide, silent plain that people on Earth call the **Sea of Tranquility**. It is called a sea, but there is not a single drop of water there.

On the way there were six craters, some big and some small, scattered about like upturned bowls. Cuội stepped close to the edge of the first one; the dust crumbled with a soft crackle beneath his feet, and he quickly stepped back.

He went around this crater and avoided that one, his eyes always on the point of light. The closer it got, the more it sparkled: sometimes flaring up white as a star, sometimes vanishing as if playing hide-and-seek.

> ***[Illustration suggestion]*** *Cuội walks along the rims of craters across a flat grey plain; far away, a small point of light blinks on the ground.*

Past the last crater, Cuội stopped short. In front of him was a small square panel, tilted on a low stand. Its face held lots of little glass blocks in rows, each one catching the light and tossing it back, sparkling like a handful of stars that had fallen to the ground. The panel had no eyes, no mouth, no arms or legs.

- Hey! Caught you!

A dazzling ray of light flashed straight into Cuội's face. He shielded his eyes with his hand and laughed.

- Who are you, being so playful?
- I'm **Twinkle**! My full name is very long: Laser Ranging Retroreflector, LRRR for short, which means the laser retroreflector panel of Apollo 11. But that name sounds so serious, I like Twinkle better. Every time I speak, my face flashes a little. Can you see?
- I'm Cuội. I'm looking for Hằng, and I need friends to light up this star lantern.

Cuội held up the star lantern, which now had one point lit gold while the rest were still dark. Twinkle flashed once, very quickly.

- Oh, Round Eye has already lit one point for you! I'll help, but first you have to play a riddle game with me. Listen: I don't know how to glow, and I don't shoot out any light myself. Yet I make light come back. What am I?
- A… mirror?
- Close! Only half right. Guess the other half and I'll tell you everything. For now, here's a prize for being close.

Twinkle flashed, and a picture appeared in the air. In it, Twinkle himself stood upright on the ground, his face turned up to the sky, standing neat and straight as if he'd just been cleaned.

- This is me when everything was done: opened out, standing straight, looking up at the sky. But now look at me…

Cuội looked again. The panel's face had grown dull with dust, and the little glass blocks barely glimmered.

> ***[Illustration suggestion]*** *Twinkle standing upright on the Moon in the picture floating in the air; beside it, the real Twinkle covered in dust, his glass face dim.*

· · ·

Cuội Wipes the Glass

- I'm covered in dust, Cuội. Light has a hard time getting in and out. Would you wipe me clean?

Cuội nodded. He took the hem of his shirt and wiped, one square at a time. The grey dust came away under his hand, showing the little glass blocks clear as water. Each time one area lit up, Twinkle burst into giggles.

When the last square shone bright, a second picture appeared on the glass, like an old image lying under the dust for years: a flat box lying on the ground, not yet opened, still folded.

- This is when they had just set me down — Twinkle said, his voice very small. — Back then I was still folded, like a book not yet opened. I couldn't do anything, just lie there and wait.
- Wait for what?
- For someone to open me and aim me the right way. You see, my work can only be done when I face Earth. Turn a little off, and the light gets lost.

> ***[Illustration suggestion]*** *A flat box lying on Moon dust, still folded, in an old yellowed picture frame; Cuội slowly wipes the dust off the glass.*

• • •

The Two-Mirror Test

- Now it's your turn to see how I work! — Twinkle cried.

Twinkle shot out a thin ray of light, and two small mirrors appeared in the air. The first mirror stood still. The second one Cuội could turn.

- Your task: turn the second mirror so that the ray hits mirror one, hits mirror two, and then slips through that narrow slit.

Cuội tried turning it. The ray bounced off and slid past the slit. He turned it back, nudged it a little, then a little more. He tried and tried until the two mirrors stood at right angles to each other, like two walls in the corner of a room.

And then the ray flew in, touched both mirrors, turned around and flew back, going right into the narrow slit.

- That's it! When two mirrors are at right angles, the light goes back out the way it came in. That's my secret! — Twinkle cried. — But don't turn it crooked: if it's off by just one degree, at 89 or 91 degrees, the light slips past the slit.
- But do you really only have two mirrors like these?
- No. These two mirrors are just a model to make it easy to play. The real me has three faces at right angles to each other, like the inside corner of a box, and I have lots of them. That way, light is sent back to wherever it came from.

> ***[Illustration suggestion]*** *Two small mirrors at right angles; a ray of light flies in, hits mirror one, hits mirror two, then flies back out through a narrow slit.*

• • •

How Long Does a Hello Take?

- Now let me tell you what I do every day — Twinkle said. — On Earth there are people who gaze at the Moon, and they send up here a very strong flash of light, called a laser pulse. I don't send anything myself, I just catch it and send it back along the same path.
- And then?
- Then they start a clock. From the moment it's sent to the moment the light comes back, however long that takes tells them the distance from them to me.

A clock appeared in the air, its hand ticking little by little. When the hand stopped, it showed **2.56 seconds**.

- That's there and back — Twinkle said. — Light is very fast, it travels three hundred thousand kilometres every second. Can you work out how far I am from Earth?

Cuội took a small stick and wrote in the dust: light took 2.56 seconds, but that was there and back, so one way is only half. He wrote each line carefully.

300,000 × 2.56 ÷ 2 = 384,000 km

- Three hundred and eighty-four thousand kilometres! — Cuội gasped.
- That's right. I rounded this number to make it easy to calculate; in real life it changes a little as the Moon moves along its orbit. But see: one small flash of light can measure such a long distance. That is Earth's hello to me, and I say hello back.

• • •

Two Pictures to Remember

Twinkle flashed, light as breathing. Two pictures appeared side by side in the air.

- Now help me put them in the right order. Which comes first, which comes second?

Cuội looked at the two pictures and put the still-folded box first, and the upright mirror panel second.

- That's right! — Twinkle said cheerfully. — Let me tell you. First, the Apollo 11 astronauts set me down on the ground, and I was still folded. After that, they opened me up and aimed me the right way. That's why that was the first time I truly got to work.

Twinkle paused for a moment, his face glowing softly.

- I am Twinkle, the mirror panel of Apollo 11. I was placed in the Sea of Tranquility in 1969. I have lots of little glass blocks. I need no battery and no motor; just turn me the right way and I'll send light back to you. I have been remembered.

When Twinkle finished, Cuội and Twinkle said together, very softly:

> **Please don't forget my name.**
>
> **I remember you now. I will tell your story.**

The star lantern in Cuội's hands trembled gently. A tiny ray of light flew from the panel into the lantern, and the **second point of the star** lit up, glowing like silver.

> ***[Illustration suggestion]*** *Cuội holds up the star lantern; two points are lit, gold and silver, shining on Twinkle's sparkling mirror panel.*

- Thank you, Cuội — Twinkle said. — You know, a small piece of equipment like me can still help people understand very great distances. Because you remembered me, I feel useful again.

Cuội smiled. He was about to say goodbye when Twinkle flashed once and pointed to the right edge of the plain.

- Look there. The grey dust is rippling in soft rings, like water when someone touches it lightly. A friend of mine is there, a very good listener. Go and meet him, Cuội. And remember to tell him my story.

Cuội nodded. He waved goodbye to Twinkle and walked toward the soft rings on the grey dust, in the sparkling light of a moonlit night.

Chapter 3: Sharp Ear

Faint ripples spread across the grey dust, one ring after another, as if someone far away were tapping softly. Cuội followed them, walking slowly and lightly, because he felt that making noise here would be rude.

Soon the ripples stopped at a flat, grey, quiet patch of ground. In the middle stood three objects, set well apart, each one different. One was tall, with an antenna on its head like a little umbrella. One stretched out three long arms, like the legs of a spider. The last was low and round, wearing a silver cloak that spread out like a skirt.

> ***[Illustration suggestion]*** *Cuội stands in the middle of the grey ground, looking at three pieces of equipment set apart from each other: a tall one with an antenna, one with three arms, and a low one covered in silver; faint ripples still spread across the dust.*

Cuội raised the star lantern. Two points of the star were already lit, one gold and one silver, but the other three were still dark.

- Hello. Is anyone here? — Cuội whispered.

There was no answer, only a few more ripples spreading softly across the dust. Then a small, hesitant voice, as if afraid of startling someone, said:

- Could you… speak quietly, please? I hear better when everything around me is very still.
- Sorry! — Cuội lowered his voice. — I'm Cuội. Who are you? There are three things here, and I don't know which one you are.
- Have a good look and guess. I get a little shy when people mix me up.

· · ·

Three Objects: Which Is the Friend?

Cuội walked around each one. When he came close to the tall one with the antenna, a small line of writing lit up on the ground: “Central station: receives commands from Earth and sends data back home.”

Cuội went up to the one with three arms. A second line lit up: “Magnetometer: measures the magnetic field around it.”

Cuội stepped up to the low, round one in the silver cloak. A third line appeared, smaller and thinner than the first two: “Seismometer: measures very tiny movements of the ground.”

- It's you! — Cuội cried, then quickly covered his mouth.
- That's right — said the small voice, as if letting out a breath. — I am Sharp Ear. My full name is PSE, which means “Passive Seismic Experiment”. That's hard to remember, isn't it?
- What do you do?
- I measure whether the ground shakes, even a tiny bit, so tiny that nobody could feel it.
- But the Moon has no air to carry sound. — Cuội was puzzled.
- You're right. So I don't hear with ears. I feel the ground shaking. I like to call it “listening”, that's all.

Sharp Ear fell quiet for a moment. Then a picture appeared in the air: a low, round machine in a silver cloak, standing on grey ground where it had just been set down.

- This is me, on 19 November 1969. An astronaut named Alan Bean took this picture. The silver cloak helps keep me from getting too hot or too cold. The nights here are very cold.

> ***[Illustration suggestion]*** *Sharp Ear's memory: a low machine in a silver cover, just set down on grey ground; far away, the shadow of an astronaut.*

· · ·

The Mixed-Up Cards

- I have a notebook where I wrote down every time the ground shook — Sharp Ear said. — But it was so long ago that the note cards got mixed up. Could you sort them out for me?

Three cards appeared, each with a wavy line and a short note. The first card: “A shake from inside the Moon, then it fades.” The second card: “Something from outside falls, and then the ground shakes.” The third card: “Regular beats, like the ticking of a

machine.” Three labels lay beside them: **Moonquake**, **Meteoroid impact** and **Noise to check**.

Cuội read each card carefully. At first he put the third card under “Moonquake”, but Sharp Ear coughed softly. He read it again and moved it.

- Even beats like a machine… I suspect those come from me — Sharp Ear said quietly. — Near sunrise and sunset, people saw me record odd disturbances, and they suspected the cover or the cable was stretching and shrinking with the heat and cold. They weren't sure, but it should be checked first.
- So looking at the wavy line isn't enough?
- That's right. You have to read what comes with it too. And by the way, the waves on these cards are ones I drew to make this easier. They aren't my real data.

When the three cards were in place, a second picture appeared. In it, Sharp Ear stood in front, and behind him the central station was being set up.

- That day, my friends and I were set down side by side — Sharp Ear said. — I don't talk much, but it's warm to have friends beside me.

> ***[Illustration suggestion]*** *The second picture in the air: Sharp Ear in front, the tall central station with its antenna behind, being set up on grey ground.*

· · ·

Why Does the Moon “Ring”?

- There's one day I remember best, Cuội — Sharp Ear said. — The day after I was set down: 20 November 1969.

Two drawn lines appeared in the air, both on one time scale from 0 to 60 minutes. The first shot up and faded very fast. The second grew until about the seventh or eighth minute, then dropped very, very slowly, and still left a trace after the fifty-fifth minute.

- That day, after the astronauts had left, people let the ascent stage of the Apollo 12 spacecraft fall onto the Moon. I recorded the shaking. Can you guess which line is mine?

Cuội studied the two lines for a long time, then pointed to the second. Sharp Ear flickered softly, as if smiling.

- Then guess how long the ground kept shaking. A few seconds, more than fifty-five minutes, or many days?
- More than fifty-five minutes! — Cuội answered.
- Correct. Some records say more than fifty-five minutes, some say about fifty, but either way it's a very long time.
- Why so long? Is the Moon like a bell?
- No. The Moon isn't hollow like a bell. The crushed rock scatters the shaking in every direction, and it fades away very slowly. So the ground keeps shaking. People say the Moon “rings” to make it easier to picture, but it isn't the sound of a bell.

> ***[Illustration suggestion]*** *Two lines side by side on the time scale; one fades fast, one rises and then softens very slowly, stretching past the fifty-fifth minute.*

· · ·

A Pattern to Open the Memory

- One more thing, Cuội — Sharp Ear said. — My memories are kept behind a stone slab. The slab only opens when you remember the right rhythm.

A grey stone slab rose out of the ground. On it were three rows of glowing beats: the first row had three even beats, the second had two beats close together, a long pause, then one more, and the third had only two beats. Beside them was a sample rhythm that Sharp Ear wanted Cuội to remember.

Cuội watched the sample once, then twice, then three times. Two beats close together, a long pause, one more. He chose the second row.

The slab went quiet for a moment, then slowly parted and opened. Inside appeared the last picture: a machine with three arms in front and, behind it, an astronaut named Pete Conrad setting up an antenna.

- My three-armed friend was here that day too — Sharp Ear said. — Every time I look at this picture, I remember I was never alone.

• • •

Sharp Ear's Notebook

The three pictures lay in Cuội's hands. Sharp Ear said softly:

- Put them in the right order for me, please. So my story doesn't get mixed up.

Cuội arranged them: first, Sharp Ear in his silver cloak, because that was when he had just been set down. Next, Sharp Ear beside the central station. Last, the three-armed machine with the person setting up the antenna.

Sharp Ear read each page of the notebook, his voice small and steady.

- I am PSE, and everyone calls me Sharp Ear. I was set down in the region called the Ocean of Storms on 19 November 1969, as part of the ALSEP equipment set of the Apollo 12 mission. I measure very small shakings of the ground. I also recorded the crash of the ascent stage. I worked until 30 September 1977, and then people switched the equipment off. But what I recorded is still kept on Earth, so scientists can still read it.

When Sharp Ear finished, Cuội and Sharp Ear said together, very softly:

Please don't forget my name.

I remember you now. I will tell your story.

The star lantern in Cuội's hands trembled very gently. A tiny ray of light flew out from Sharp Ear's silver cloak, circled Cuội and slipped into the lantern. The third point of the star lit up, a soft blue like moonlight, beside the gold and the silver.

> ***[Illustration suggestion]*** *Cuội holds up the star lantern; three points are lit, gold, silver and soft blue, casting light on the grey ground; Sharp Ear in his silver cloak stands beside him.*

- Thank you for being willing to listen, Cuội — Sharp Ear said. — Most people only remember loud things. Small measurements like mine, hardly anyone

notices. But thanks to small things like these, people learned more about a place very far away.

Cuội smiled. He was about to say goodbye when Sharp Ear flickered softly and pointed to the right edge of the ground.

- Look there. A tiny line of light is stretching across the dust, like a silver thread. Please follow it.

Cuội didn't know where the line of light led. But in the story of the Moon, a small light always knows the way to friends who are waiting.

Cuội nodded. He waved goodbye to Sharp Ear, very softly, and walked on after the line of light, further and further away, in the gentle glow of a moonlit night.

Chapter 4: Mesh Wheels

Cuội followed the small line of light. It stretched over gentle slopes, grey boulders and patches of dust as fine as flour. He walked so far that telling it all would take longer than one night. All you need to know is that, after a very long way, Cuội came out onto a wide plain, with great mountains on one side and a long, winding groove in the ground on the other.

The line of light led straight to a small vehicle, parked silently in the dust. It had four wheels woven from metal threads, a light frame, two empty seats and many straps. It had no eyes and no mouth. But as Cuội came close, the star lantern in his hands trembled.

> ***[Illustration suggestion]*** *Cuội stands in front of a vehicle with four metal mesh wheels, parked on a wide plain with mountains far off to one side; a faint line of light stretches behind him; the three-point star lantern glows in his hands.*

- Oh, a visitor! I thought you were only here to look — a warm, quick voice said.
- I'm Cuội. And you are…?
- My name is Lunar Roving Vehicle, which means a car for exploring the Moon. But please call me Mesh Wheels, because I'm very proud of these four wheels.
- Who drives you?
- Two astronauts: David Scott and James Irwin. I don't drive myself, you know, I'm not a robot. I carried them around to explore this place in late July and early August 1971, on the Apollo 15 mission. Their lander was called Falcon. I was the first vehicle like me to go along with astronauts.
- How can I help you?
- Would you help me get ready for a trip? I'm starting to forget what I rode on.

· · ·

Choosing the Wheels for the Trip

Mesh Wheels set three kinds of wheels on the ground: a rubber wheel with treads, a metal mesh wheel, and a solid, stiff metal wheel.

Cuội rolled each one across a small strip of ground with a rock on it. The rubber wheel bounced, then sank a little. The stiff metal wheel lifted its whole axle when it hit the rock. The mesh wheel bent, hugged the rock, and rolled over it smoothly.

- I choose the mesh wheel — Cuội said.
- Correct! A metal mesh wheel can bend, and it has little cleats to grip loose soil. But don't think rubber is bad. Another Apollo mission used rubber tires on a small cart. Which wheel is best depends on the job.
- But testing wheels in a drawing isn't a real measurement, is it?
- Right. It's only a way to help you see the differences.

Mesh Wheels flashed, and a picture appeared in the air. In it, an astronaut was driving Mesh Wheels across grey ground, with mountains far behind.

- This is when astronaut Scott drove me back to the Falcon lander, with rocks and soil on board. Then they climbed aboard and flew home. And I stayed behind.

> ***[Illustration suggestion]*** *Mesh Wheels' memory: an astronaut riding the four-mesh-wheeled vehicle back to the Falcon lander, carrying sample bags; far mountain ridges behind.*

· · ·

Drawing the Way Out and the Way Back

- Now help me remember the route — Mesh Wheels said. — Going far is exciting, but what matters more is remembering the way back.

A simple map appeared on the ground, divided into squares. There was a starting point, the lander. There were two stops, marked A and B. There were deep pits that must not be entered, and a few slopes that use more battery to climb.

- I only have this much battery — Mesh Wheels pointed at a small battery bar. — But that's not all. Remember this: however far you go, if anything goes wrong, the two people on me must still be able to walk back to the lander.

Cuội squinted at the map, then drew with his finger in the dust. He went from the lander to stop A, around to stop B, avoided the pits, and came back. In one place Cuội changed

direction because the way would have meant climbing a slope. In another he rubbed out his line and drew it again.

- This route takes eight steps and nine battery units. The farthest point is still only four walking steps from the lander — Cuội said.
- Right! — Mesh Wheels cheered. — It doesn't have to be the shortest, as long as you can go and come back. By the way, the twelve battery units and four steps are numbers I made up to keep the game easy, not numbers from the real flight.

> ***[Illustration suggestion]*** *A square-grid map drawn in the dust; a route joins the lander, stop A and stop B, then returns, avoiding dark pits and crossing two small slopes.*

• • •

A Sampling Trip

Then Mesh Wheels said, in a voice full of secrets:

- Now I'll take you on a real trip. Hop aboard. This is an imaginary trip, and I'll carry you along the very route you just drew.

Cuội sat on the empty seat. Mesh Wheels rolled slowly forward along the route Cuội had drawn, across the squares, over the slopes, avoiding the deep pits. At stop A, the vehicle stopped.

At stop A there was a big rock. Cuội took a small hammer. If he tapped too lightly, the rock didn't budge. If he hit too hard, the rock flew away. He had to hit just right. After two good taps, the rock cracked and a piece came off. Cuội picked up the piece, carried it back to the vehicle and put it in the bag.

At stop B, Cuội saw three different patches of soil. He raised a magnifying glass and looked closely at each one. Two had only crumbled grey rock. The one in the middle was stranger: its grains were finer, mixed with small broken pieces.

- This soil is so different! — Cuội gasped.
- You have sharp eyes — Mesh Wheels said. — Scientists like to compare patches of soil like these, because each one tells a story about the place where it lies.

Cuội scooped up a little of the middle soil, carried it back to the vehicle and stored it carefully beside the rock.

- The Apollo 15 astronauts brought back more than 77 kilograms of real rock and soil samples to Earth for scientists to study — Mesh Wheels said. — The way you took samples today is a simpler version I made for the game, but it's true that I carried them out to collect samples.

Cuội and Mesh Wheels went back to the parking place. Mesh Wheels flashed again, and a second picture appeared: a photo taken from very high up, looking down over a whole stretch of land. In the middle was a tiny dot, the base of the Falcon lander that had stayed behind, and further to the right was me. From it, faint marks stretched out like threads.

- A friend who flies very high took this picture, in 2011 — Mesh Wheels said. — My tracks are still there.

> ***[Illustration suggestion]*** *A photo taken from high above, looking down on the land: the small dot is the base of the lander, faint marks stretch out like threads; Cuội and Mesh Wheels look up at the picture together.*

• • •

Mesh Wheels' Diary

Two pictures lay in Cuội's hands. Mesh Wheels said:

- Put them in the right order for me, please. So my story doesn't get mixed up.

Cuội arranged them: the 1971 picture first, because that was when Mesh Wheels was still working with the astronauts. The 2011 picture second, because that was when people looked back at that place from high above.

Mesh Wheels read each page, his voice slow and warm.

- I am Mesh Wheels, the lunar roving vehicle of the Apollo 15 mission. In late July 1971, I carried David Scott and James Irwin to explore the Hadley–Apennine region. When they went home, I stayed. I learned one thing: if you go far, you must remember the way back. And because someone remembers me, my story is not forgotten.

When Mesh Wheels finished, Cuội and Mesh Wheels said together, very softly:

> **Please don't forget my name.**
>
> **I remember you now. I will tell your story.**

The star lantern in Cuội's hands shook. A tiny ray of light flew out of the four mesh wheels, circled Cuội and slipped into the lantern. The fourth point of the star lit up, a warm copper glow.

> ***[Illustration suggestion]*** *Cuội holds up the star lantern; four points are lit, gold, silver, soft blue and warm copper, casting light on the four-wheeled vehicle parked at the foot of the mountain.*

- Thank you, Cuội — Mesh Wheels said. — I stay here, but my story doesn't stay here alone any more.

Cuội was about to say goodbye when Mesh Wheels looked up at the black sky.

- Look there. A little light is passing across the sky, steady as a night watchman. That's a friend of mine, who flies very high. He's still awake, and he can see all the places I've been.

Cuội looked up. In the black sky, a tiny point of light was quietly travelling from one side to the other. At his feet, a faint line of light appeared on the dust, pointing to the right edge.

Cuội waved goodbye to Mesh Wheels. He walked on after the line of light, looking back once more at the four-wheeled vehicle standing quietly at the foot of the mountain, in the silver light of a moonlit night.

Chapter 5: Sky Watcher

Cuội followed the line of light across the dust. The farther he went, the blacker the sky grew and the more stars there were. The small point of light above kept passing from one side to the other, steady and patient, never stopping.

Cuội stopped and held the star lantern high. Four points of the star were lit, and only one was dark. Suddenly the lantern trembled, and a voice came out of it, clear and thin as the night wind, with a few soft crackles mixed in:

- Hello. I can see you from up here.

> ***[Illustration suggestion]*** *Cuội stands alone on the grey plain, holding up the star lantern; in the black sky above, a small point of light glides by, and a thin thread of light joins it to the lantern.*

- Oh! Where are you? — Cuội asked, looking up.
- I'm very high up, flying around the Moon. I can't come down to you, so I talk to you through your lantern. My full name is Lunar Reconnaissance Orbiter, or LRO for short. But please call me Sky Watcher.
- What do you do up there?
- I take pictures of the Moon. I have cameras called LROC that look down at the surface, helping scientists study the land and the marks left in places people have visited.
- Can you see me?
- I can see mountains, craters and long tracks. But small things aren't clear. It's like sitting on a plane: you can see a whole city, but not every brick. Looking from far away is useful, but it has limits too.
- I'm looking for Hằng. Can you help me?
- I will. But first, help me remember the places I've seen from up high.

Matching Memories to Places

Two pictures taken from high above appeared in the air, like two small maps. Each showed a different region.

- Tell me which region your friends live in — Sky Watcher said.

Cuội thought of Round Eye. He pointed to the Ocean of Storms, where the Apollo 12 mission had landed. Sky Watcher flashed, and a picture appeared: an astronaut standing beside a three-legged machine, Surveyor 3 itself.

- That's right. In November 1969, two Apollo 12 astronauts visited him and took home a camera and some parts to study. I saw that place from above, and I took pictures of it.
- Sharp Ear is in that region too — Cuội added. — He was set down in the Ocean of Storms.
- Yes. The two of them are in the same region.

Cuội thought of Twinkle. He pointed to the Sea of Tranquility, where the Apollo 11 mission had landed. Sky Watcher flashed again. Another picture appeared, with small notes naming the area.

- That's right. The mirror panel there sends laser light back to where it came from. But don't think I can see every single mirror. From up high, I can only see the whole area.
- I've seen where Mesh Wheels drove, too. His tracks are still there — Sky Watcher added.

> ***[Illustration suggestion]*** *Two pictures taken from high above, one of the Ocean of Storms and one of the Sea of Tranquility; Cuội touches each one, and a smaller picture appears inside each.*

• • •

A New Spot in the Picture

- There's one more story I want to tell, and I'd like you to help me find something — Sky Watcher said. — Look at these two pictures.

Two pictures appeared side by side, both of the same piece of land. The left one was taken before 12 February 2012. The right one was taken after 28 July 2013. Both had three numbered regions: 1, 2 and 3.

- There's a place in the right picture that isn't in the left one. See if you can find it.

Cuội looked closely at each region. He picked region one, but it looked exactly like the left picture. He picked region three, the same. Then he looked at the middle region: there was a small bright spot that wasn't in the earlier picture at all.

- Region two! — Cuội cried.
- Exactly. It's a new crater, about eighteen metres wide. On 17 March 2013, a rock from space struck the Moon. Telescopes on Earth saw a flash of light. Afterwards, I photographed that piece of land again, and I found this crater.
- So the Moon is still changing.
- Yes. One more thing: the day this crater formed was 17 March, not the day I took the pictures. Please remember the difference.

> ***[Illustration suggestion]*** *Two pictures side by side, three numbered regions; in the right-hand picture, region two has a small bright spot circled; high above, Sky Watcher glides by.*

· · ·

Keeping Watch Over the Dark

- There's one more thing — Sky Watcher said. — My watch is usually near the Moon's south pole, where the Sun always stays low on the horizon. There are some very dark places there.

A drawing appeared: a cross-section of a crater, with a ledge on each side and a deep floor in the middle. Above it was a slider for changing the direction of the sunlight.

- Slide the sunlight to each direction, then see how the left ledge, the deep floor and the right ledge change. Then choose the spot that is always dark.

Cuội slid the slider to one side. The left ledge lit up, but the crater floor stayed dark. He slid it to the middle: both ledges grew bright, and the floor stayed dark. He slid it to the

right: the right ledge was bright, the left ledge went dark, and the floor stayed pitch black.

- The deep floor! — Cuội said. — Whichever way the sun shines, the crater wall still blocks it.
- Write it in my watch book. But please remember: this is a drawing to make things easy to understand. To be sure about a real place on the Moon, scientists must use a great many pictures and a lot of measurements of the terrain.

> ***[Illustration suggestion]*** *A cross-section of a crater with a slider for the sun's direction; three small pictures show the left and right ledges lit differently while the crater floor stays dark; a watch book lies open beside it.*

• • •

Three Pages of the Watch Book

Sky Watcher paused for a moment, then said:

- Now I'll tell you the three pages in my watch book. Listen, then help me remember them.

The first page appeared: a picture of the Apollo 12 area with notes, and in it the small dot of Surveyor 3. “Places you know, seen from high above,” Sky Watcher said. “From far away, I can recognise a place, but I can't see small things clearly.”

The second page appeared: the picture after the 2013 impact, with the bright spot circled. “The Moon still changes,” Sky Watcher said. “A flash of light seen from Earth, and a crater I found in a picture.”

The third page appeared: a map of light near the south pole, put together from a great many pictures. “A watch over the dark,” Sky Watcher said. “No single picture can tell the whole story. You have to look closely, compare, and check again and again.”

When Sky Watcher finished, Cuội and Sky Watcher said together, very softly:

Please don't forget my name.

I remember you now. I will tell your story.

The star lantern in Cuội's hands trembled. From far, very far away, a thin thread of light drifted down from the point of light above, wound around Cuội once and slipped into the lantern. The fifth point of the star lit up, a clear white like starlight. Now all five points shone together.

> ***[Illustration suggestion]*** *Cuội holds up the star lantern; all five points are lit, gold, silver, soft blue, copper and clear white, casting light across the grey plain; high above, Sky Watcher glides by.*

- Thank you, Cuội — Sky Watcher said. — From up here I can see a great deal, but there are things only someone standing on the ground can see. Thanks to you, my story is complete.
- Now let me show you the way to Hằng.
- Hằng isn't far away at all — Sky Watcher said. — Go back to where you began. I'll light your way.

From the point of light above, a long path of light spread down to the ground and ran toward the horizon. Far away, in the places Cuội had passed, four small lights blinked: Round Eye, Twinkle, Sharp Ear and Mesh Wheels. Each sent out a tiny ray that greeted Cuội and joined the path of light, as if his friends were standing up to see him home.

Cuội smiled. He waved to the point of light above, then walked along the road of light toward the banyan where he had begun.

Part 3: The Light Returns

Cuội returned to the banyan. Behind him came his five friends: Round Eye, Twinkle, Sharp Ear, Mesh Wheels and, circling high above, Sky Watcher. The star lantern in his hands shone with all **five points**.

The lantern suddenly felt weightless. It left Cuội's hands, rose high, and opened, pouring light over a wide patch of ground around the banyan. There, little flowerbeds sprang up, each carrying a name: one for Round Eye, one for Twinkle, one for Sharp Ear, one for Mesh Wheels, and one for Sky Watcher. This was the **Garden of Memory**.

> ***[Illustration suggestion]*** *The Garden of Memory beneath the banyan: five glowing flowerbeds, each with a friend's name; the star lantern floats above.*

Then a familiar scent returned: cinnamon blossoms. Soft music began to play. From the middle of the light, a gentle figure appeared.

- Hằng! — Cuội cried.

Hằng smiled and walked to the stone seat.

- I never went anywhere, Cuội. But when everyone forgets one another, my light grows dim, so dim that you couldn't see me. You lit each point of the star, which means you remembered each friend. And that is why I could appear again.

Cuội's eyes filled with tears. Hằng took his hand and pointed to the sky.

Far, far away hung a tiny blue ball: the Earth. Around it shimmered a ring of light. Looking closer, Cuội saw that these were not stars at all, but countless small fragments drifting by.

- That is what people have left behind — Hằng said softly. — Some are friends, and some are fragments that nobody remembers by name.

The banyan swayed its leaves as if nodding. The friends stood around, listening in silence. Then Hằng spoke, slowly, so that every word would settle in the listeners' hearts:

- Science grows every day, Cuội. I am so glad to see people dare to fly far, dare to ask and dare to learn.
- But science must not be paid for by leaving a swarm of rubbish wandering around our shared homes: the Earth, the Moon and other worlds.
- Let science walk together with caring for the environment: not only for your own roof, but also for the friends all around you.

Back on the Porch

On the porch, the star lantern in little An's hands suddenly glowed bright.

- Grandma! I see them! Hằng is sitting beside Cuội!

Grandma smiled, said nothing, and only held An's hand tightly.

> ***[Illustration suggestion]*** *Little An and Grandma on the porch; the lantern glows bright, and on the Moon two small figures sit side by side under the banyan.*

That night, An wrote her very first promise in her little notebook: “Every day, I will do one small thing to look after the friends around me.”

And high above, where the ancient banyan still stands, the five points of the star kept shining, for ever after.

Afterword

Cuội's story ends here, but yours has not ended yet.

On the Moon, the metal friends still lie there, quiet, and as long as someone remembers their names, they are no longer lonely. Around the Earth, and in many other places, the things people have made are still waiting for a simple question: **how far will we remember, and how far will we take responsibility?**

Science is wonderful. But the most beautiful science walks together with care: care for our shared home, care for the friends around us, and care even for the machines that worked for us.

Three small things you can do today

1. Name something you are using, and think about where it will go when you no longer need it.
2. Pick up one small piece of rubbish and put it in the right place.
3. Tell someone close to you about a friend you met in this book.

My Promise

Today, I promise to:

_____

_____

_____

_____

My name: _____ Date: ____ / ____ / _____

Reader's Notebook

This section is for readers who want to know more. Here are the **real** things about each friend in the story, with pictures of those objects so you can explore together. Keep exploring, young readers!

Did you know? Round Eye

Round Eye in the story is NASA's **Surveyor 3** spacecraft, an **uncrewed robot** launched on an Atlas-Centaur rocket on 17 April 1967 to find out whether the Moon's surface was firm enough for later landings with people on board.

- Surveyor 3 landed in a region called Oceanus Procellarum (“Ocean of Storms”), though there is no water there. On landing, it **bounced twice** before it came to rest.
- Its camera sent about **6,326 pictures** back to Earth.
- Its arm with a small scoop was used to **dig trenches and press on the soil**, helping scientists measure how firm Moon soil is.
- In November 1969, astronauts **Pete Conrad and Alan Bean** of Apollo 12 visited Surveyor 3 and brought its camera and some other parts back to Earth for study. The rest of the spacecraft stayed on the Moon.

Real or story?

In the story	Real or story?
Round Eye talks and feels sad	**Story.** Surveyor 3 is a machine; we imagined his voice.
Surveyor 3 bounced twice when landing	**Real.**
Round Eye took more than six thousand pictures	**Real.** About 6,326 pictures.
Two astronauts visited in November 1969	**Real.** They were Pete Conrad and Alan Bean (Apollo 12).

In the story	Real or story?
Round Eye was waiting for Cuội	**Story.**

[Image placeholder]
Figure 1: A footpad of Surveyor 3 and its landing marks

Image note: *Insert a photo or drawing of the footpad. If using a NASA image, credit "NASA" and check its usage terms.*

[Image placeholder]
Figure 2: Trenches dug by Surveyor 3

Image note: *Photographed in 1969 but showing trenches dug in 1967; state both years in the caption.*

[Image placeholder]
Figure 3: Apollo 12 astronauts beside Surveyor 3

Image note: *Insert a photo or drawing. Credit "NASA" and check its usage terms.*

Did you know? Twinkle

Twinkle in the story is the **Laser Ranging Retroreflector (LRRR) of Apollo 11, a laser retroreflector panel that belongs to the EASEP equipment set, placed** in the **Sea of Tranquility** on the Moon in **1969**. It is a very special piece of equipment: it can't talk and doesn't make its own light, but it helps people measure the distance to the Moon.

- The panel is made of many fused-silica cubes, each one a corner-cube prism. Each prism has faces at right angles to each other, like the inside corner of a box, so light that comes in from any direction is sent back in exactly that direction.
- The panel **needs no battery or motor** to reflect light, and it **does not produce laser light itself**.
- Observatories on Earth send **laser pulses** to the Moon, then measure how long the light takes to go and come back to work out the one-way distance. The light's round trip takes a little over two and a half seconds.
- The distance to the Moon changes along its orbit. The figures **2.56 seconds** and **384,000 km** in the story are **rounded** to make the sum easy.
- The Apollo 11 astronauts set the equipment down on the Moon’s surface; it has a folding support so that it can be aimed toward Earth.

Real or story?

In the story	Real or story?
Twinkle talks, plays riddles and laughs	**Story.** The panel is equipment; we imagine his voice and playful nature.
Cuội wipes the dust off Twinkle	**Story.** Cuội is a character in the story; the dust and the wiping are pictures in the story.
The two mirrors Cuội turns to test	**Model.** Only to explain things simply. The real equipment has prisms with three faces at right angles.
The astronauts set it down, open it and aim it	**Real.** The two pictures in the notebook are NASA photographs.

In the story	Real or story?
2.56 seconds and 384,000 km	**Rounded numbers.** In real life, the distance changes with the Moon's orbit.

[Image placeholder]
Figure 1: The equipment just set down, still folded (AS11-40-5945)

Image note: *NASA image AS11-40-5945: the equipment just set down. Credit "NASA", the image code, and check the usage terms for the image.*

[Image placeholder]
Figure 2: The equipment opened and aimed (AS11-40-5952)

Image note: *NASA image AS11-40-5952: the equipment opened and aimed. Credit "NASA" and the image code. Do not guess how much dust is on the device from how light or dark the photo looks.*

[Image placeholder]
Figure 3: Diagram of light reflecting inside a prism

Image note: *Draw an illustrative diagram: the ray enters, hits the faces at right angles, and returns in exactly the direction it came. Label clearly that this is an illustration, not a real photo.*

• • •

Did you know? Sharp Ear

Sharp Ear in the story is the **PSE (Passive Seismic Experiment),** an experiment that uses seismometers and was part of the ALSEP science equipment set of the **Apollo 12 mission.** It was set down in the region called **Oceanus Procellarum** ("Ocean of Storms") on 19 November 1969.

- The PSE measures **very tiny movements of the ground**. It doesn't "hear" with ears, and the Moon has no air to carry sound.
- The story has three separate pieces of equipment: the **PSE** (the seismometer), the **central station** that receives commands and sends data back to Earth, and a **magnetometer** with three arms.
- The ground on the Moon can shake because of a **moonquake** (coming from inside) or because something from outside crashes into it. To work out what caused a shaking, scientists need to read the information that comes with it, not just look at the wavy line.
- On 20 November 1969, after the astronauts had left, the **ascent stage** of Apollo 12 was made to crash onto the Moon. The PSE recorded shaking that grew stronger until about the seventh or eighth minute and then faded slowly. The shaking lasted about 50 minutes.
- The shaking lasted so long because it was **scattered** in the layer of crushed rock and lost its energy very slowly, not because the Moon is hollow like a bell. The word "ringing" is only a way of making it easier to picture.
- Near sunrise and sunset, the PSE recorded odd disturbances. People suspected the cover or the cable was stretching and shrinking, but that was only a suspicion, not a firm conclusion.

- ALSEP stopped working on 30 September 1977. The Apollo data is still kept so that scientists can study it.

Real or story?

In the story	Real or story?
Sharp Ear talks, is shy and loves to listen	**Story.** The PSE is measuring equipment; we imagine his voice and personality.
Cuội sorts the mixed-up cards	**Story.** The note cards and the three wavy lines were drawn for the story to keep the game easy; they are not real NASA data.
Sharp Ear measures the shaking of the ground	**Real.** The PSE is a seismometer that measures very tiny movements of the ground.
Two lines compared on a 0–60 minute scale	**Model.** Only to explain things simply; they are not NASA's raw graphs.
The ascent stage crashing onto the Moon on 20 November 1969	**Real.** The ground shook for about 50 minutes.
The stone slab that opens when the right rhythm is remembered	**Story.** The rhythm and the slab were made up for the story.
Three pictures taken by Alan Bean	**Real.** The three pictures in the notebook are NASA photographs, taken in 1969.
A small line of light leading onward	**Story.** The line of light is a sign leading to the next friend, made up for the story.

[Image placeholder]
Figure 1: Sharp Ear (the PSE) in its silver cover (AS12-47-6916)

Image note: *NASA image AS12-47-6916, taken by Alan Bean on 19 November 1969: the PSE and its thermal cover. Credit "NASA · Alan Bean", give the image code, and check the usage terms.*

[Image placeholder]
Figure 2: The PSE and central station being set up (AS12-47-6918)

Image note: *NASA image AS12-47-6918: the PSE in front, the central station behind, being set up. Do not caption it as showing the antenna already installed. Credit "NASA · Alan Bean" and give the image code.*

[Image placeholder]
Figure 3: The three-armed magnetometer (AS12-47-6921)

Image note: *NASA image AS12-47-6921: the magnetometer in front, Pete Conrad setting up an antenna behind. Credit "NASA · Alan Bean" and give the image code.*

• • •

Did you know? Mesh Wheels

Mesh Wheels in the story is the **Lunar Roving Vehicle (LRV)** of the **Apollo 15 mission,** the vehicle driven by astronauts David Scott and James Irwin as they explored the Hadley–Apennine region during three moonwalks, from 31 July to 2 August 1971. It was a vehicle with a driver, not a self-driving robot, and Apollo 15 was the first mission to use this kind of vehicle. Their lander was named Falcon.

- The LRV's tires were made of woven steel strands, with V-shaped titanium strips covering about half of the contact area to grip loose soil.
- The wheel test in the story is about **this vehicle**. Not every device on the Moon used mesh wheels: the Apollo 14 mission used rubber tires on a small cart.

- The vehicle ran on two single-use batteries, weighed about 210 kilograms, and travelled about 28 kilometres in total over the three moonwalks.
- The vehicle was designed by NASA's Marshall Space Flight Center and built by Boeing; Apollo 16 and Apollo 17 also each carried one.
- The Apollo 15 astronauts really did bring back more than 77 kilograms of rock and soil, including a core sample from about 3 metres deep. The way samples are collected in the story has been simplified.
- In 2011, the orbiting LRO spacecraft photographed the area from about 25 kilometres up, and in the picture the vehicle, the base of the Falcon lander and the wheel tracks can still be seen.

Real or story?

In the story	Real or story?
Mesh Wheels talks and tells stories	**Story.** The LRV is a vehicle with a driver; it has no eyes, mouth, arms or legs. The voice is something we imagine.
Wheels woven from metal	**Real.** The LRV used flexible metal mesh wheels.
Cuội tests three kinds of wheel on a strip of ground	**Model.** This is an illustration for observing, not a measurement of speed or strength.
A square map, 12 battery units and a limit of 4 walking steps	**Model.** Set by the story to keep the game easy, not numbers from the real flight.
Cuội rides along and collects samples with Mesh Wheels	**Story.** This is an imaginary trip, explained before the game.
Collecting rock and soil	**Partly real.** The astronauts really did collect rock and soil samples, but the way it's done in the story is simplified.
Mesh Wheels remembers driving back to Falcon	**Partly real.** A NASA photo shows astronaut Scott driving the vehicle back to the Falcon lander with rock and soil; Mesh Wheels' telling is story.
A light that lights the fourth point of the star	**Story.** The star lantern and the friends' light are made up for the story.

[Image placeholder]
Figure 1: Astronaut David Scott on the lunar roving vehicle, heading back to Falcon (NASA/MSFC)

Image note: *NASA image published on the page "Lunar Roving Vehicle, Apollo 15 Mission" (nasa.gov), credit NASA/MSFC. The source page gives no image code or date taken; do not add any.*

[Image placeholder]
Figure 2: A close-up of the metal mesh wheel

Image note: *An illustration of a mesh wheel showing the woven threads, open gaps and V-shaped ridges. If you crop the image, say clearly that it is a crop from a NASA image; do not alter the image content with AI.*

[Image placeholder]
Figure 3: The vehicle's tracks seen from orbit (LROC M175252641L,R)

Image note: *NASA LROC image M175252641L,R, taken in 2011 from about 25 km up, credit NASA/GSFC/Arizona State University. Keep the labels already in the image. The source page gives no month taken: say only 2011, and do not use the publication date (5 March 2012) as the date taken.*

• • •

Did you know? Sky Watcher

Sky Watcher in the story is the **Lunar Reconnaissance Orbiter (LRO),** a spacecraft that watches the Moon from orbit; it was launched on 18 June 2009 on an **Atlas V 401 rocket.** Its **LROC** cameras photograph the Moon's surface, helping scientists study the terrain and the marks left in places people have visited.

- Pictures taken from orbit **are not close-ups**. From high above we can see terrain and large marks clearly, but not small objects.
- A picture of the whole Apollo 11 area can't be used to say we have seen the laser reflector. The picture only shows that area from far away.
- On 17 March 2013, a rock from space struck the Moon. A telescope on Earth recorded a **flash of light**. Afterwards, LRO's pictures helped find the **new crater**, about 18 metres wide.
- The two pictures in the story were taken before 12 February 2012 and after 28 July 2013. The day of the impact (17 March 2013) is a different day, so don't mix it up with the days the pictures were taken.
- Maps of light near the Moon's south pole are made by combining many LROC pictures. The drawing of three sun directions in the story is only an **educational diagram**; it is not enough to prove that a real spot on the Moon is dark forever.
- In the story, Cuội is always standing on the ground when he talks to Sky Watcher. The scene in orbit is an illustration, not to scale.

Real or story?

In the story	Real or story?
Sky Watcher talks to Cuội through the lantern	**Story.** LRO is an observing spacecraft; the voice and the lantern are made up for the story.
Round Eye in the Ocean of Storms, Twinkle in the Sea of Tranquility	**Real.** Surveyor 3 and Apollo 12 are in the Ocean of Storms, and Apollo 11 is in the Sea of Tranquility.
Sky Watcher takes pictures from high above	**Real.** LRO flies around the Moon and takes pictures with its LROC cameras.

In the story	Real or story?
Two pictures from before and after the impact	**Real.** The before-and-after pair from 2013 are NASA pictures. The three numbered regions were added by the story.
A new crater about 18 metres wide	**Real.** According to NASA material.
A slider for the sun's direction, three regions to watch	**Model.** Just a teaching diagram, not a lighting simulation of a real crater.
The “deep floor” is always dark	**Model.** This is the game's answer; scientific conclusions need many pictures and measurements.
The path of light pointing to the banyan	**Story.** The map to Hằng is made up for the story.

[Image placeholder]
Figure 1: The Moon-observing spacecraft LRO

Image note: *Insert a photo or illustration of the LRO spacecraft. If using a NASA image, credit “NASA” and check its usage terms.*

[Image placeholder]
Figure 2: The before-and-after picture pair of the 2013 impact (NASA SVS 4242)

Image note: *The earlier picture taken 12 February 2012 and the later picture taken 28 July 2013, with the new crater outlined. State the impact date, 17 March 2013, separately. Credit NASA SVS 4242.*

[Image placeholder]
Figure 3: A map of light near the Moon's south pole (NASA SVS 4716)

Image note: *An illumination map of the south pole put together from many LROC pictures. Credit NASA SVS 4716 and make clear that it is a reference image, not a conclusion about any permanently dark place.*

Little Dictionary

Word	Simple meaning
ALSEP	A set of science equipment that Apollo astronauts left on the Moon to take measurements over a long time.
Antenna	A part used to receive or send radio signals.
Apollo 11	NASA's first crewed mission to land on the Moon, in 1969.
Apollo 12	NASA's crewed mission to the Moon in November 1969.
Apollo 15	NASA's crewed mission to the Moon in late July 1971, the first to carry a lunar roving vehicle.
Ascent stage	The part of a lander used to lift astronauts off the Moon.
Astronaut	A person trained to travel in space.
Corner-cube prism	A block of glass with faces at right angles like the corner of a box, sending light straight back the way it came.
Crater	A round hollow in the ground, usually made by an impact.
Hadley–Apennine	A region of the Moon at the foot of the Apennine mountains, where the Apollo 15 mission landed.
Lander	A spacecraft built to touch down on a world such as the Moon.
Laser	A very strong, very straight beam of light that travels very far without spreading out.
Laser pulse	A very short flash of laser, used to measure the time to go and return.

Word	Simple meaning
Laser ranging retroreflector (LRRR)	Equipment on the Moon with many prisms that sends laser light straight back the way it came.
LRO	Lunar Reconnaissance Orbiter, a spacecraft that flies around the Moon and takes pictures of its surface.
Lunar roving vehicle	A vehicle with a driver, used to travel farther on the Moon and to carry tools and samples.
Magnetic field	The space around a magnet where there is a pull or push on things that react to magnets.
Mesh wheel	A wheel woven from metal threads, able to bend as it rolls over rocks and loose soil.
Meteoroid	A rock from space that falls onto a world.
Moon dust	Fine, crumbled rock and soil covering the Moon's surface.
Moonquake	A shaking of the ground on the Moon, which can start from deep inside the Moon.
NASA	The United States' space and aeronautics agency.
Noise	An unwanted signal mixed into a recording that makes the results hard to read.
Observatory	A place on Earth where scientists use telescopes and instruments to watch the sky.
Orbit	The path of flight around a world, like a spacecraft circling the Moon.
Reflection	When light hits a surface and bounces back.
Rock sample	A piece of rock or soil taken back so scientists can study it.
Round trip	Going and coming back. The round-trip time of light is the time light takes to go and return.
Scattering	When waves or light are spread in many directions by small stones or specks of dust.
Sea of Tranquility	A flat, dark plain on the Moon. It is called a sea but has no water.
Seismometer	A machine that measures very tiny movements of the ground.
Speed of light	Light travels about 300,000 kilometres every second.

Word	Simple meaning
Uncrewed robot	A machine controlled from far away or working on its own, with no person on board.

Questions to Talk About

Grown-ups and young readers can try answering these together. There are no right or wrong answers.

1. If you were Round Eye, would you be scared bouncing twice when landing? Why?

2. Why does Round Eye want people to remember his name?

3. Twinkle needs no battery or motor and still does its job. What do you think is useful about that, so far from Earth?

4. Light travelling from Earth to the Moon and back takes just over two seconds. What else in life "goes and comes back" like that (an echo, a bouncing ball…)?

5. Sharp Ear has no ears, yet he can "listen" to the Moon. What other ways can you think of to find out something your ears can't hear?

6. Sharp Ear is small and shy, but does important work. Do you know anyone or anything small that is useful?

7. If you had to travel very far, what would you prepare so that you could always find your way back?

8. Mesh Wheels stayed on the Moon after his last trip. Why should people remember the names of the machines that helped them?

9. From high above, Sky Watcher sees things you can't see from the ground, but some things are hard to see from far away. Have you ever seen something that looks different from far away and from close up?

10. Sky Watcher says that to be sure, you have to compare many pictures. Why isn't looking just once always enough?

11. Is there an old object at home whose name and story you would like to remember?

12. What do you think happens if we don't take responsibility for what we make?
13. What will you do to look after our shared home?

Sources and AI Used

Surveyor 3 – launched by Atlas-LV3C Centaur-D

- https://science.nasa.gov/mission/surveyor-3/
- https://science.nasa.gov/mission/surveyor-3/stories/
- https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1967-035A
- https://nssdc.gsfc.nasa.gov/nmc/spacecraft/displayDataset.action?spacecraftId=1967-035A

LRRR – Laser Ranging Retroreflector (Apollo 11)

- https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1969-059C
- https://nssdc.gsfc.nasa.gov/nmc/experiment/display.action?id=1969-059C-04
- https://nssdc.gsfc.nasa.gov/nmc/dataset/display.action?id=PSCM-00011
- https://ilrs.gsfc.nasa.gov/docs/2026/lunar/Apollo11_Preliminary_Sci_Report_LLR.pdf

PSE – Passive Seismic Experiment (Apollo 12)

- https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1969-099C
- https://nssdc.gsfc.nasa.gov/nmc/experiment/display.action?id=1969-099C-03
- https://nssdc.gsfc.nasa.gov/nmc/dataset/query.action?spacecraftName=Apollo+12+Lunar+Module+%2F+ALSEP&experimentName=Passive+Seismic+Experiment+%28PSE%29

LRV – Lunar Roving Vehicle (Apollo 15)

- https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1971-063C
- https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo_lrv.html
- https://www.nasa.gov/image-article/lunar-roving-vehicle-apollo-15-mission/

LRO – Lunar Reconnaissance Orbiter – launched by Atlas V 401

- https://science.nasa.gov/mission/lro/
- https://svs.gsfc.nasa.gov/gallery/lunar-reconnaissance-orbiter/
- https://science.nasa.gov/mission/lro/about/

AI Used

- Claude AI – script and story support
- SunoAI – audio support
- OpenAI Codex – game development support

Acknowledgements

Thank you to NASA for publishing open material about its Moon missions. Thank you to the organizers of the NASA International Space Apps Challenge, who gave us the chance to tell this story. Thank you to the Mid-Autumn storytellers who keep Cuội, Hằng and the banyan alive in our childhoods. And thank you, reader, for reading, and for remembering the names of the forgotten friends.

Hanoi, 15 November 2026

The authors – Light of memories